



Statistics for Data Science

Lesson 3 — Sampling, Correlation & Covariance

Why We Sample • Sampling Methods • Bias • Sample Size • Covariance • Correlation • Correlation vs Causation

Why Do We Sample?

We almost never study an entire population — here is why

Impossible

You cannot survey every person on Earth, test every product ever made, or measure every transaction in history. The population is simply too large.

Too Slow

By the time you finished collecting data from everyone, the information would be outdated. Samples allow decisions to be made quickly.

Too Expensive

Studying every member of a population would cost enormous amounts of money and resources. A well-chosen sample gives the same insight at a fraction of the cost.

Good Enough

A properly selected sample of 1,000 people can accurately represent a population of 10 million. Statistics gives us tools to measure exactly how accurate it is.

Types of Sampling

How you select your sample determines how trustworthy your results are



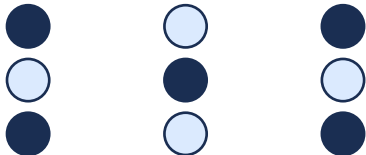
Simple Random Sampling

Every member of the population has an equal chance of being selected. Selection is completely random — like drawing names from a hat.

Pros: Unbiased. Easy to understand.

Cons: May miss important subgroups if they are small.

Select 500 random customer IDs from a database of 100,000.



Stratified Sampling

Divide the population into subgroups (strata) based on a characteristic, then randomly sample from each stratum proportionally.

Pros: Ensures all subgroups are represented.

Cons: Requires knowing the population structure in advance.

Survey 100 from each age group: 18-30, 31-50, 51+.



Systematic Sampling

Select every k-th member from a list. Start at a random point, then pick every k-th item after that.

Pros: Simple to implement on ordered lists.

Cons: Can introduce bias if the list has a hidden pattern.

From a sorted customer list, select every 10th record.



Sampling Bias

When your sample does not truly represent the population



What is Sampling Bias?

Sampling bias occurs when some members of the population are more likely to be selected than others, causing the sample to not truly represent everyone.

The result: your conclusions will be wrong — not because of bad analysis, but because of bad data collection.

Bias is one of the most serious problems in data science because it is often invisible and can mislead decisions at scale.

Selection Bias

The method of selection systematically favors certain groups.

Example: Surveying customers only through email excludes those without internet access.

Survivorship Bias

You only see the 'survivors' — the failures are invisible.

Example: Studying successful startups to learn what makes a company great, ignoring all the ones that failed with the same strategies.

Voluntary Response Bias

People who respond voluntarily tend to have stronger opinions.

Example: Online ratings where mostly very satisfied or very angry customers leave reviews.

How to avoid: Use random selection, ensure all groups are reachable, document your sampling method

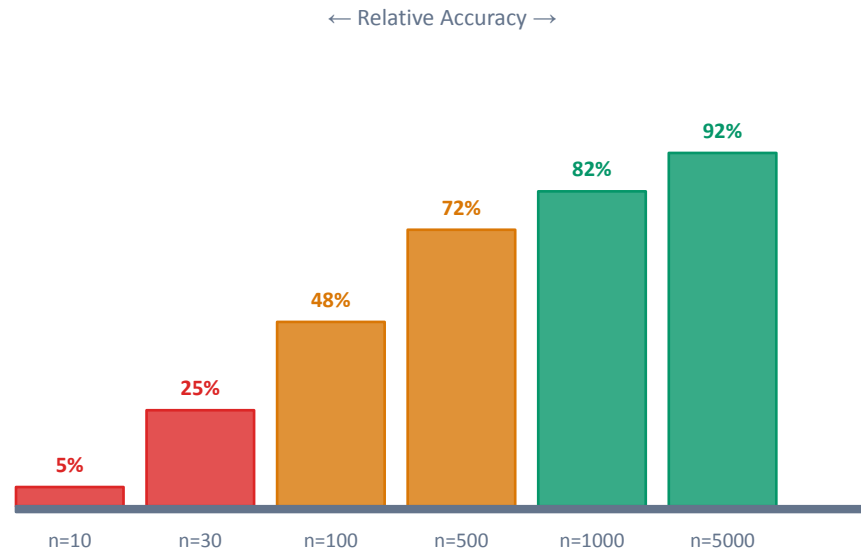
Sample Size — How Large is Large Enough?

Bigger samples give more accurate estimates, but there are diminishing returns

Key Principles

- 1 Larger samples → more accurate estimates → smaller margin of error
- 2 $n \geq 30$ is the minimum for the Central Limit Theorem to apply
- 3 The relationship is not linear — going from $n=100$ to $n=200$ helps much more than going from $n=10,000$ to $n=10,100$
- 4 Sample size depends on: required confidence level, acceptable margin of error, and how varied the population is
- 5 A sample of 1,000 can represent a population of 1 million almost as well as 10 million — population size matters less than you think

Accuracy vs Sample Size



Notice: gains slow down rapidly after $n=500$

What is Covariance?

Measures whether two variables tend to move in the same direction

Definition

$$\text{Cov}(X, Y) = \sum (x_i - \bar{x})(y_i - \bar{y}) / n$$

Covariance tells you:

Positive covariance: when X increases, Y tends to increase too.

Negative covariance: when X increases, Y tends to decrease.

Zero covariance: no linear relationship between X and Y.

The problem: the value depends on the units of measurement. A covariance of 500 could be large or tiny depending on scale. This makes it hard to interpret directly.

Visual Examples



Positive Covariance



Negative Covariance



Near-Zero Covariance

What is Correlation?

Covariance standardized — now the scale does not matter

Pearson Correlation (r)

$$r = \text{Cov}(X, Y) / (\sigma_x \times \sigma_y)$$

$$r = +1$$

Perfect positive linear relationship

$$r = -1$$

Perfect negative linear relationship

$$r = 0$$

No linear relationship at all

$$|r| > 0.7$$

Strong correlation

$$0.3-0.7$$

Moderate correlation

$$|r| < 0.3$$

Weak correlation

Correlation Spectrum



$$r \approx +0.9$$

Strong positive



$$r \approx +0.4$$

Moderate positive



$$r \approx 0$$

No relationship



$$r \approx -0.9$$

Strong negative

Pearson vs Spearman Correlation

Two different ways to measure correlation — for different types of data

	Pearson (r)	Spearman (r _s)
What it measures	Linear relationship between two continuous variables	Monotonic relationship — one variable tends to increase when the other does, even if not linearly
Data type	Continuous numerical data	Ranked data, ordinal data, or continuous data with outliers
Assumes	Both variables are normally distributed. No major outliers.	No distributional assumption. More robust to outliers.
Formula	$r = \text{Cov}(X,Y) / (\sigma_x \times \sigma_y)$	$r_s = \text{Pearson } r \text{ applied to the ranks of the data}$
When to use	Height vs weight. Temperature vs sales. Salary vs experience.	Customer satisfaction rank vs repurchase rate. Income vs happiness rating.

Correlation vs Causation

One of the most important distinctions in data science

Correlation does NOT imply causation.

Just because two things move together does not mean one causes the other.

What Correlation Tells You

Two variables move together — when one goes up, the other tends to go up (or down) too.

This is useful for prediction: knowing X helps you estimate Y.

But you cannot say which one CAUSED the other.

Three Possible Explanations

1. A causes B

Smoking causes cancer. (Confirmed by experiments.)

2. B causes A

Reversed direction. Being sick causes low activity, not the other way.

3. A third variable C causes both

Ice cream sales and drowning rates are correlated. Both are caused by summer (hot weather).

Spurious Correlation

Real correlation, zero meaning — a trap that fools even experienced analysts

What is Spurious Correlation?

A spurious correlation is a statistical relationship between two variables that appears real — sometimes very strong — but has no meaningful connection.

It happens because:

- Both variables happen to trend over time
- A hidden third variable influences both
- Pure coincidence in small datasets
- The analyst went looking for patterns (data dredging)

In data science, spurious correlations are dangerous because they can look just like real ones in your analysis. Always ask: is there a logical mechanism that could explain this?

Correlation: $r = 0.95$

Nicolas Cage films per year ↔ Drownings in swimming pools

Both happened to trend similarly over the same years. No causal link whatsoever.

Correlation: $r = 0.87$

Per capita cheese consumption ↔ Deaths by bedsheet tangling

A famous example from Tyler Vigen's Spurious Correlations. Pure coincidence over time.

Correlation: $r = 0.78$

Number of data science job posts ↔ Global avocado sales

Both grew strongly in the same period. Both are driven by broader economic trends, not each

Rule: Always ask — is there a plausible mechanism? Can an experiment confirm it?

Lesson 3 — Key Takeaways



We sample because studying an entire population is usually impossible, too expensive, or too slow



Correlation standardizes covariance to a value between -1 and $+1$ — scale-independent



Random sampling gives equal chance. Stratified ensures subgroups are represented. Systematic picks every k -th item.



$r = +1$ is perfect positive. $r = -1$ is perfect negative. $r = 0$ means no linear relationship.



Sampling bias occurs when selection favors certain groups — results become unreliable



Pearson assumes linearity and normality. Spearman uses ranks — more robust to outliers.



Larger samples reduce error, but gains diminish rapidly. $n \geq 30$ is usually the minimum.



Correlation does NOT imply causation — both could be caused by a third variable



Covariance measures direction of relationship between two variables — but the value depends on scale



Spurious correlations look real but have no meaningful connection — always ask for a mechanism